

Pandas

Lecture1

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Lecture1 Content



- Pandas Introduction
- Pandas Series
- Pandas DataFrame
- Read Data
 - CSV
 - XML
 - JSON
 - Excel
 - HTML
 - SQL
- Write Data
 - CSV
 - SQL
- Elementary operations
- Compute some basic statistics
- References

Pandas

- Pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language
- Python with *pandas* is in use in a wide variety of academic and commercial domains, including Finance, Neuroscience, Economics, Statistics, Advertising, Web Analytics, and more



Pandas Series

	First Name
0	Lois
1	Brenda
2	Joe
3	Diane
4	Benjamin
5	Patrick
6	Nancy
7	Carol
8	Frances
9	Diana



PANDAS
SERIES

Pandas DataFrame

The diagram illustrates a Pandas DataFrame with 7 rows and 6 columns. The columns are labeled *Name*, *Team*, *Number*, *Position*, and *Age*. The rows are indexed from 0 to 6. Annotations include: 'Columns' with arrows pointing to the column headers; 'Rows' with arrows pointing to the row indices; and 'Data' with a box highlighting specific cells (Jonas Jerebko, 8.0, NaN, PG, NaN) and arrows pointing to them from the label.

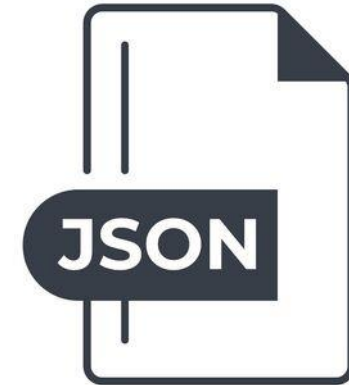
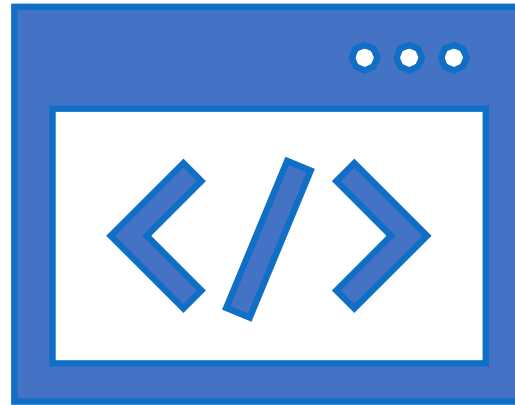
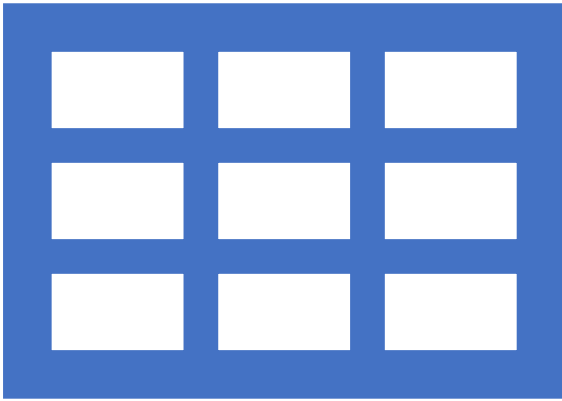
	<i>Name</i>	<i>Team</i>	<i>Number</i>	<i>Position</i>	<i>Age</i>
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

Rows

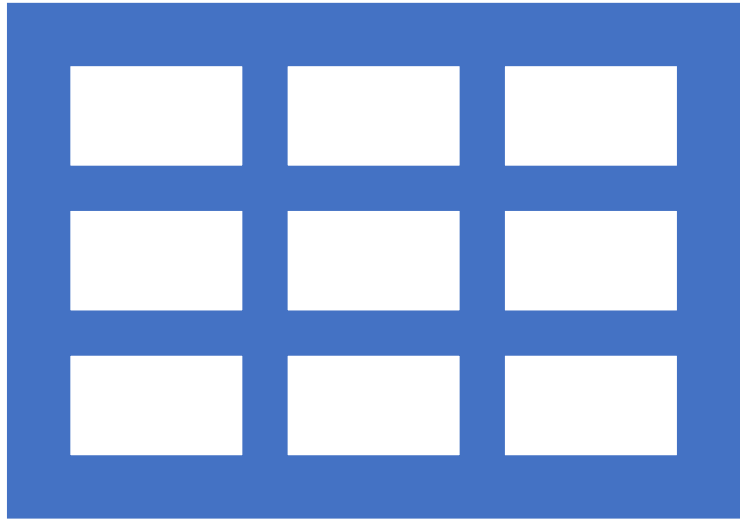
Columns

Data

Read Data



Write Data

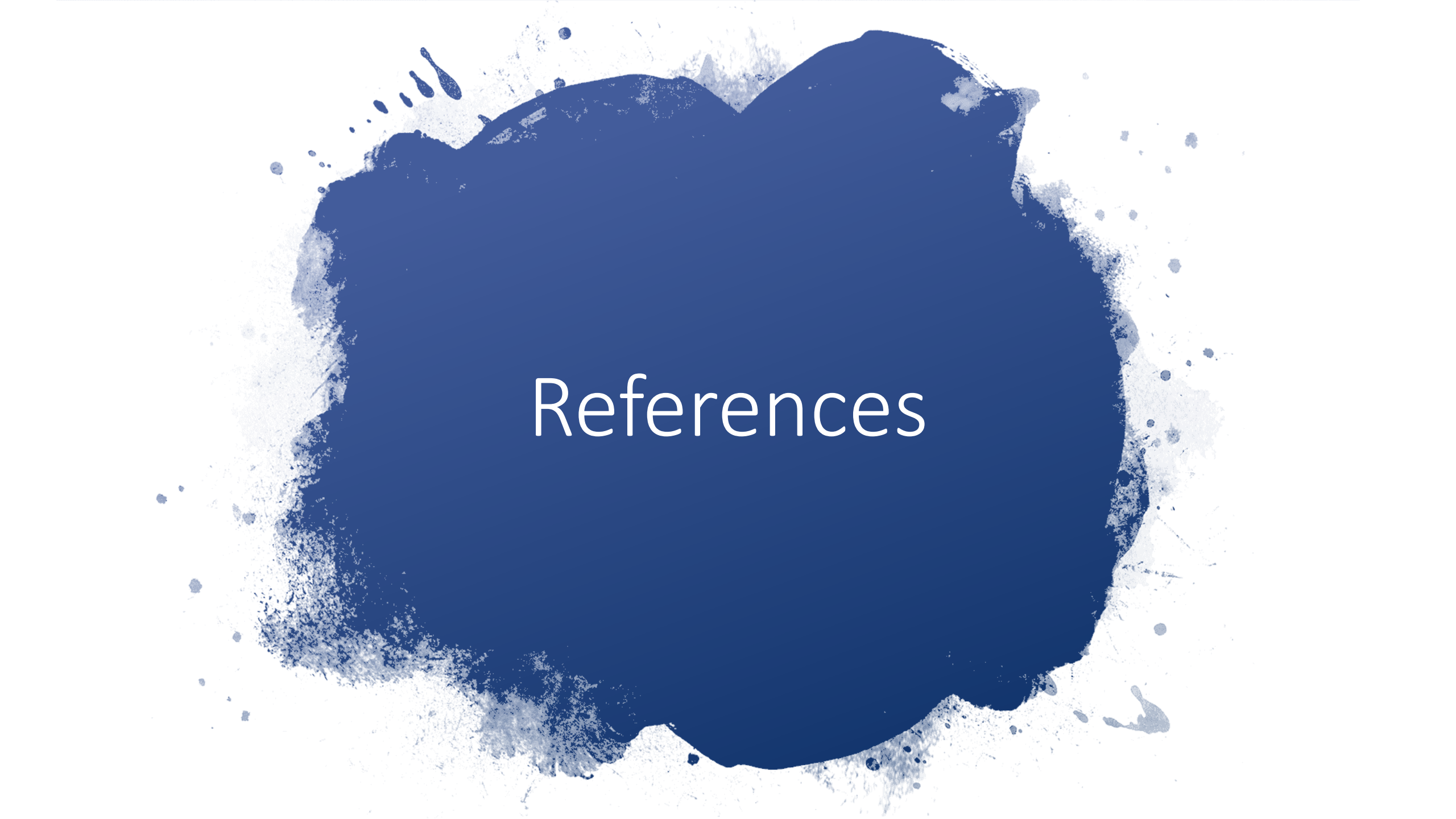


Another Practical Demos



- Elementary operations
- Compute some basic statistics



A large, dark blue ink splatter or blotch is centered on a white background. The splatter has irregular, feathered edges with some smaller droplets and speckles trailing off from the main mass. The word "References" is written in white, sans-serif font in the center of the dark blue area.

References

References

- <https://pandas.pydata.org/>
- <https://www.linkedin.com/learning/pandas-for-data-science/>



thank you